STRENGTH OF MATERIALS LAB

(23A01301P)

B.Tech. Civil Engineering - II Year I Semester - JNTUA R23

L-T-P-C : 0-0-3-1.5

Course Objectives

- To determine the tensile strength and yield parameters of mild steel.
- To find the flexural strengths of steel/wood specimens and measure deflections.
- To determine the torsion parameters of mild steel bar.
- To determine hardness numbers, impact and shear strengths of metals.
- To determine the load-deflection parameters for springs.

Course Outcomes

- Conduct tensile test and draw stress-strain diagrams for ductile metals.
- Perform bending test and determine load-deflection curve of steel/wood.
- Conduct torsion test and determine torsion parameters.
- Perform hardness, impact and shear tests and calculate hardness numbers, impact and shear strengths.
- Conduct tests on closely coiled and open coiled springs and calculate deflections.

List of Experiments

1. Tension test
2. Bending test on cantilever beam (steel/wood)
3. Bending test on simply supported beam
4. Torsion test
5. Hardness test
6. Compression test on open coiled springs
7. Tension test on closely coiled springs
8. Compression test on wood / concrete
9. Izod / Charpy impact test on metals
10. Shear test on metals
11. Use of electrical resistance strain gauges
12. Continuous beam - deflection test

Experiment 1: Tension Test

Aim:

To conduct a tension test on a mild steel specimen and determine yield stress, ultimate stress, breaking stress, % elongation, % reduction in area and Young's modulus; to draw the stress-strain curve.

Apparatus:

Universal Testing Machine (UTM), mild steel specimen, vernier calipers, micrometer, extensometer, punch and scale.

Formulae / Theory:

Original area A0 = (pi/4) d^2 (d = original diameter)
Yield stress = Load at yield / A0
Ultimate stress = Maximum load / A0
Breaking stress = Load at fracture / Final area
% Elongation = (Lf - L0)/L0 x 100
% Reduction in area = (A0 - Af)/A0 x 100
Young's modulus E = Stress / Strain (in the elastic region)

Procedure:

1. Measure the diameter of the specimen at three places and take the mean; mark the gauge length L0.
2. Fix the specimen firmly in the upper and lower jaws of the UTM.
3. Attach the extensometer and set the load and elongation scales to zero.
4. Apply the load gradually and note the elongation at regular load increments.
5. Note the load at yield point (drop of pointer), the maximum load and the breaking load.
6. Remove the broken pieces, join them, measure the final gauge length and the neck diameter.
7. Plot the stress-strain curve and calculate the required values.

Observations:

Original diameter d = 12.5 mm, gauge length = 62.5 mm, A0 = 122.7 mm^2. Yield load = 31.0 kN, maximum load = 47.5 kN, breaking load = 36.2 kN, final gauge length = 78.5 mm, neck diameter = 8.6 mm.

Load (kN)	Extension (mm)	Stress (N/mm^2)	Strain
5	0.013	40.7	0.000206
10	0.026	81.5	0.000416
15	0.038	122.2	0.000606
20	0.052	163.0	0.000836
25	0.064	203.7	0.001016
30	0.077	244.5	0.001232

Result:

Yield stress = 252.6 N/mm^2; Ultimate stress = 387.1 N/mm^2; Breaking stress = 623.2 N/mm^2; % Elongation = 25.6 %; % Reduction in area = 52.7 %; E = 1.98 x 10^5 N/mm^2.

Experiment 2: Bending Test on Cantilever Beam

Aim:

To determine the Young's modulus (E) and flexural stress of a steel/wood cantilever beam by measuring deflection under load.

Apparatus:

Cantilever beam setup with clamp, hanger and weights, dial gauge, vernier calipers, scale.

Formulae / Theory:

Deflection at free end: $y = W L^3 / (3 E I)$

So $E = W L^3 / (3 y I)$

Moment of inertia $I = b d^3 / 12$ (rectangular section)

Bending stress = $M y / I$ with $M = W L$

Procedure:

1. Measure the breadth b, depth d and effective length L of the beam.
2. Clamp the beam rigidly at one end to form a cantilever.
3. Place the dial gauge at the free end and set it to zero.
4. Add weights in equal steps and note the dial gauge reading each time.
5. Unload step by step and note the readings again.
6. Calculate E for each load and take the mean; plot Load vs Deflection.

Observations:

Steel beam: breadth b = 25 mm, depth d = 6 mm, effective length L = 500 mm, $I = 450\ mm^4$.

Load W (N)	Loading (mm)	Unloading (mm)	Mean y (mm)	E (N/mm^2)
2	0.90	0.93	0.91	202642
4	1.80	1.84	1.82	203224
6	2.71	2.71	2.71	205167
8	3.69	3.78	3.73	198503
10	4.53	4.55	4.54	203996

Result:

Young's modulus of the cantilever beam, $E = 2.03 \times 10^5\ N/mm^2$. The load-deflection graph is a straight line within the elastic limit.

Experiment 3: Bending Test on Simply Supported Beam

Aim:

To determine the Young's modulus and the load-deflection behaviour of a simply supported beam with a central point load.

Apparatus:

Simply supported beam apparatus with knife edges, hanger and weights, dial gauge, vernier calipers, scale.

Formulae / Theory:

Central deflection: $y = W L^3 / (48 E I)$

So $E = W L^3 / (48 y I)$

$I = b d^3 / 12$

Maximum bending moment $M = W L / 4$, Bending stress $= M / Z$

Procedure:

1. Measure the breadth, depth and span L of the beam.
2. Place the beam on the two knife-edge supports.
3. Put the dial gauge below the mid-span and set it to zero.
4. Apply loads at the centre in equal steps and note the deflection at each step.
5. Unload and note the readings; plot Load vs Deflection.
6. Calculate E for each reading and find the average.

Observations:

Steel beam: breadth b = 25 mm, depth d = 10 mm, span L = 600 mm, $I = 2083.3\ mm^4$.

Load W (N)	Loading (mm)	Unloading (mm)	Mean y (mm)	E (N/mm^2)
50	0.54	0.55	0.55	196608
100	1.08	1.07	1.08	200157
150	1.67	1.58	1.62	199863
200	2.21	2.13	2.17	199115
250	2.64	2.64	2.64	204528

Result:

Young's modulus of the beam, $E = 2.00 \times 10^5\ N/mm^2$ (average value).

Experiment 4: Torsion Test

Aim:

To determine the modulus of rigidity (G), shear stress and angle of twist of a mild steel bar under torque.

Apparatus:

Torsion testing machine, mild steel bar, vernier calipers, micrometer, scale, protractor / twist-measuring dial.

Formulae / Theory:

T / J = G x theta / L = tau / R

Modulus of rigidity G = T L / (J x theta)

Polar moment of inertia J = (pi/32) d^4

Shear stress tau = T R / J

Procedure:

1. Measure the diameter and gauge length of the specimen.
2. Fix the specimen between the two chucks of the machine.
3. Set the torque and angle-of-twist readings to zero.
4. Apply torque in steps and note the angle of twist each time.
5. Continue until the specimen fails (if required).
6. Plot Torque vs Angle of twist and calculate G.

Observations:

Diameter d = 10 mm, gauge length L = 200 mm, J = 981.7 mm^4.

Torque T (N-m)	Twist (deg)	Twist (radian)	G (N/mm^2)
5	0.74	0.0129	79081
10	1.43	0.0250	81517
15	2.23	0.0389	78467
20	2.90	0.0507	80392
25	3.62	0.0631	80667

Result:

Modulus of rigidity of the material, G = 8.00 x 10^4 N/mm^2; maximum shear stress at T = 25 N-m is 127.3 N/mm^2.

Experiment 5: Hardness Test

Aim:

To determine the hardness number of a given metal using the Brinell and Rockwell hardness tests.

Apparatus:

Brinell / Rockwell hardness testing machine, steel ball indenter (10 mm) and diamond cone indenter, specimens (mild steel, brass, aluminium), microscope.

Formulae / Theory:

Brinell Hardness Number BHN = 2P / [pi D (D - sqrt(D^2 - d^2))]

P = load (kgf), D = ball diameter (mm), d = indentation diameter (mm)

Rockwell hardness is read directly from the dial (HRB / HRC).

Procedure:

1. Clean the specimen surface and place it on the anvil.
2. Raise the specimen until it touches the indenter and apply the minor load.
3. Apply the major load for the standard time (about 15 s for steel).
4. Release the load and measure the indentation diameter using the microscope (Brinell) or read the dial (Rockwell).
5. Repeat at three different places and take the average.

Observations:

Indenter: 10 mm steel ball (Brinell); 1/16 inch ball with 100 kgf major load (Rockwell B).

Material	Load P (kgf)	Dia. d (mm)	BHN	Rockwell
Mild steel	3000	3.95	235	HRB 82
Brass	1500	3.10	194	HRB 62
Aluminium	500	2.55	96	HRB 38

Result:

BHN of mild steel = 235, brass = 194, aluminium = 96 (ball dia. 10 mm). Rockwell: mild steel = 82 HRB, brass = 62 HRB, aluminium = 38 HRB.

Experiment 6: Compression Test on Open Coiled Springs

Aim:

To determine the stiffness and modulus of rigidity of an open coiled helical spring under compression, and to find its deflection.

Apparatus:

Spring testing machine, open coiled spring, weights, vernier calipers, scale.

Formulae / Theory:

For an open coiled spring with small helix angle (alpha):

Deflection $y = 64\ W\ R^3\ n / (G\ d^4)$ (approx.)

Stiffness $k = W / y$

$G = 64\ W\ R^3\ n / (y\ d^4)$

R = mean coil radius, n = number of turns, d = wire diameter

Procedure:

1. Measure the wire diameter d, mean coil diameter D and count the turns n.
2. Place the spring on the platform of the testing machine.
3. Note the initial length of the spring.
4. Apply compressive load in steps and note the compressed length each time.
5. Calculate deflection = initial length - compressed length.
6. Calculate stiffness and modulus of rigidity; plot Load vs Deflection.

Observations:

Wire diameter d = 6 mm, mean coil radius R = 30 mm, number of turns n = 10, free length = 120 mm.

Load W (N)	Length (mm)	Deflection (mm)	Stiffness (N/mm)	G (N/mm^2)
10	118.3	1.65	6.05	80617
20	116.7	3.34	5.98	79771
30	115.1	4.87	6.16	82155
40	113.5	6.49	6.16	82171
50	111.8	8.19	6.11	81437

Result:

Stiffness of the spring k = 6.09 N/mm; modulus of rigidity $G = 8.12 \times 10^4$ N/mm^2.

Experiment 7: Tension Test on Closely Coiled Springs

Aim:

To determine the stiffness, modulus of rigidity and deflection of a closely coiled helical spring under axial tension.

Apparatus:

Spring testing machine, closely coiled spring, hanger and weights, vernier calipers, scale.

Formulae / Theory:

Deflection $y = 64\ W\ R^3\ n / (G\ d^4)$

Stiffness $k = W / y$

$G = 64\ W\ R^3\ n / (y\ d^4)$

Maximum shear stress $\tau = 16\ W\ R / (\pi\ d^3)$

Procedure:

1. Measure the wire diameter d, mean diameter D = 2R and number of turns n.
2. Suspend the spring from the upper support and note the initial length.
3. Add weights gradually to the hanger in equal steps.
4. Note the extended length for each load.
5. Deflection = extended length - initial length.
6. Calculate k and G; plot Load vs Deflection.

Observations:

Wire diameter d = 5 mm, mean coil radius R = 20 mm, number of turns n = 12, initial length = 60 mm.

Load W (N)	Length (mm)	Deflection (mm)	Stiffness (N/mm)	G (N/mm^2)
10	61.2	1.24	8.05	79143
20	62.4	2.45	8.17	80349
30	63.6	3.65	8.23	80902
40	64.9	4.94	8.10	79591
50	66.1	6.13	8.16	80225

Result:

Stiffness of the spring = 8.14 N/mm; modulus of rigidity $G = 8.00 \times 10^4$ N/mm^2; maximum shear stress at 50 N = 40.7 N/mm^2.

Experiment 8: Compression Test on Wood / Concrete

Aim:

To determine the compressive strength of wood (parallel and perpendicular to grain) and of concrete cubes.

Apparatus:

Compression Testing Machine (CTM) / UTM, wooden blocks, concrete cubes (150 mm), vernier calipers, scale.

Formulae / Theory:

Compressive strength = Maximum load / Cross-sectional area

$f_c = P / A$ (N/mm^2)

For concrete cube: A = 150 x 150 = 22500 mm^2

Procedure:

1. Measure the dimensions of the specimen and find its area.
2. Clean the platens of the machine and place the specimen centrally.
3. Apply the load gradually at a uniform rate.
4. Note the maximum load at which the specimen fails.
5. Observe the mode of failure (crushing / shear / splitting).
6. Compute the compressive strength; repeat for three specimens.

Observations:

Wood blocks 50 x 50 x 150 mm; concrete cubes 150 mm.

Specimen	Area (mm^2)	Failure load (kN)	Strength (N/mm^2)
Wood (parallel to grain)	2500	112.0	44.8
Wood (perpendicular to grain)	2500	28.5	11.4
Concrete cube	22500	690	30.7
Concrete cube	22500	705	31.3
Concrete cube	22500	682	30.3

Result:

Compressive strength of wood: parallel to grain = 44.8 N/mm^2, perpendicular to grain = 11.4 N/mm^2; concrete (average of 3 cubes) = 30.8 N/mm^2.

Experiment 9: Izod / Charpy Impact Test on Metals

Aim:

To determine the impact strength (energy absorbed) of a metal specimen using the Izod / Charpy test.

Apparatus:

Impact testing machine (pendulum type), standard notched specimens (Izod 75 x 10 x 10 mm, Charpy 55 x 10 x 10 mm, V-notch), vernier calipers.

Formulae / Theory:

Impact energy = Energy absorbed by specimen to fracture (read from dial, Joules)
Impact strength = Energy absorbed / Area under notch (J/mm^2)
Izod: specimen held as a cantilever; Charpy: specimen simply supported.

Procedure:

1. Measure the specimen dimensions and notch depth.
2. Raise the pendulum to its initial position and lock it.
3. Place the specimen in the vice (Izod) or on supports (Charpy) with the notch facing away from / towards the striker as per standard.
4. Release the pendulum so it strikes and breaks the specimen.
5. Note the energy absorbed from the dial.
6. Repeat for other specimens and calculate impact strength.

Observations:

Izod specimen 75 x 10 x 10 mm with 2 mm V-notch; area under notch = 10 x 8 = 80 mm^2; pendulum energy = 168 J.

Specimen	Initial energy (J)	Energy absorbed (J)	Area (mm^2)	Strength (J/mm^2)
Mild steel	168	62	80	0.775
Brass	168	34	80	0.425
Aluminium	168	18	80	0.225
Cast iron	168	5	80	0.062

Result:

Impact strength: mild steel = 0.775 J/mm^2, brass = 0.425 J/mm^2, aluminium = 0.225 J/mm^2, cast iron = 0.062 J/mm^2 (Izod). Mild steel is the toughest and cast iron the most brittle.

Experiment 10: Shear Test on Metals

Aim:

To determine the shear strength of a metal (single and double shear) using the UTM.

Apparatus:

UTM with shear test attachment, mild steel round bar specimens, vernier calipers.

Formulae / Theory:

Single shear: tau = P / A

Double shear: tau = P / 2A

$A = (\pi/4)\, d^2$ (cross-sectional area of the bar)

Procedure:

1. Measure the diameter of the specimen and find the area.
2. Place the specimen in the shear attachment (single or double shear arrangement).
3. Fix the attachment in the UTM and set load to zero.
4. Apply the load gradually until the specimen shears.
5. Note the maximum load at failure.
6. Calculate the shear strength for both cases.

Observations:

Cross-sectional area A = 113.1 mm^2.

Type	Dia. d (mm)	Load P (kN)	Shear strength (N/mm^2)
Single shear	12	41.5	366.9
Single shear	12	42.3	374.0
Double shear	12	83.2	367.8
Double shear	12	84.6	374.0

Result:

Shear strength of mild steel: single shear = 370.5 N/mm^2; double shear = 370.9 N/mm^2 (average values).

Experiment 11: Use of Electrical Resistance Strain Gauges

Aim:

To measure strain in a loaded member using electrical resistance strain gauges and to determine the stress and Young's modulus.

Apparatus:

Strain gauges (foil type), Wheatstone bridge circuit / strain indicator, cantilever or tension specimen, loading weights, adhesive, connecting wires.

Formulae / Theory:

Gauge factor GF = (dR / R) / strain

Strain e = (dR / R) / GF

Stress = E x e; Bending stress = M y / I

Bridge output is proportional to strain.

Procedure:

1. Clean the surface of the specimen and bond the strain gauge with adhesive.
2. Connect the gauge to the strain indicator (quarter / half bridge) and balance the bridge.
3. Apply loads in steps and note the strain indicator reading each time.
4. Calculate stress from the applied load and section properties.
5. Compute E = Stress / Strain and plot Load vs Strain.

Observations:

Steel cantilever 25 x 6 mm, gauge at 300 mm from the load point, gauge factor = 2.1, quarter-bridge. Stress = M y / I = 2.0 x W N/mm^2.

Load (N)	Strain (microstrain)	Stress (N/mm^2)	E (N/mm^2)
5	49	10.0	202432
10	102	20.0	196529
15	152	30.0	197640
20	197	40.0	203119
25	251	50.0	199111

Result:

Strain varies linearly with load; Young's modulus from strain gauge readings, E = 2.00 x 10^5 N/mm^2. (Strain in microstrain.)

Experiment 12: Continuous Beam - Deflection Test

Aim:

To determine the deflection of a continuous beam (two spans) under point loads and compare it with theoretical values.

Apparatus:

Continuous beam apparatus with three supports, hanger and weights, dial gauges, vernier calipers, scale.

Formulae / Theory:

Theoretical deflection for a two-span beam with central load on a span is obtained from the three-moment (Clapeyron) equation.

Support moment M_B: $M_A L1 + 2 M_B (L1 + L2) + M_C L2 = - 6 A1 x1 / L1 - 6 A2 x2 / L2$

E I is taken from the beam section: $I = b d^3 / 12$

Procedure:

1. Measure the spans, breadth and depth of the beam and fix it on three supports.
2. Place dial gauges under the loading points and set them to zero.
3. Apply the load on one span in steps.
4. Note the dial gauge readings at each load.
5. Calculate the theoretical deflection and compare it with the experimental one.
6. Plot Load vs Deflection.

Observations:

Two equal spans of 400 mm each, steel section 25 x 8 mm, $I = 1066.7\ mm^4$, load applied at mid of one span. Theoretical deflection under load = $0.01497\ W\ L^3 / (E\ I)$.

Load (N)	Exp. deflection (mm)	Theo. deflection (mm)	% Difference
50	0.23	0.22	0.3
100	0.47	0.45	3.8
150	0.69	0.67	2.3
200	0.88	0.90	2.1
250	1.18	1.12	4.8

Result:

The experimental deflection of the continuous beam agrees closely with the theoretical value (average difference = 2.6 %).